

The mkt for FUCHKA has the following D eqn.

$$Q_D = 200 - 3P_F + 0.5I + 2P_C$$

where,  $Q_D$  = quantity demanded for FUCHKA

$P_F$  = price per plate of FUCHKA (in taka)

$I$  = Income

$P_C$  = price per plate of HOTPOTI (in taka)

Initial values,

$$P_F = 50, I = 5000, P_C = 60$$

2) Find initial quantity of FUCHKA. — 1 mark

$$Q_D = 200 - 3(50) + 0.5(5000) + 2(60)$$

$$\therefore Q_D = 2670 \rightarrow \text{OLD VALUE for entire mkt of QUANTITY}$$

b) Suppose, price of FUCHKA,  $P_F$ , increases to 100, while all else remains constant. Find PED. — 3 marks

$$Q_D = 200 - 3(100) + 0.5(5000) + 2(60) \quad \left. \vphantom{Q_D = 200 - 3(100) + 0.5(5000) + 2(60)} \right\} \text{new } Q$$

$$\therefore Q_D = 2520$$

$$PED = \frac{\frac{2520 - 2670}{\left[ \frac{100 + 2670}{2} \right]} \times 100\%}{\frac{100 - 50}{\left[ \frac{100 + 50}{2} \right]} \times 100\%}$$

$$\therefore PED = -0.09 \quad \left. \vphantom{\therefore PED = -0.09} \right\} \text{elasticity}$$

Interpretation:  $|PED| > 1 \rightarrow \text{ELASTIC}$   
 $|PED| < 1 \rightarrow \text{INELASTIC}$

$\hookrightarrow$  PED will always be negative due to the inverse relationship between price & demand.

FUCHKA has an inelastic demand.  $\left. \vphantom{\text{FUCHKA has an inelastic demand.}} \right\} \text{interpretation}$

c) Suppose, income has decreased to 4000, while  $P_F$  &  $P_C$  remain at their initial values.

Find YED.

$$Q_D = 200 - 3(50) + 0.5(4000) + 2(60)$$

$$\therefore Q_D = 2170$$

$$YED = \frac{\frac{2170 - 2670}{\left[ \frac{2170 + 2670}{2} \right]} \times 100\%}{\frac{4000 - 5000}{\left[ \frac{4000 + 5000}{2} \right]} \times 100\%}$$

$$YED = 0.93$$

$YED > 0$  NORMAL  $\left( \begin{matrix} I \downarrow D \downarrow \\ I \uparrow D \uparrow \end{matrix} \right)$   
 $- YED < 0$  INFERIOR  $\left( \begin{matrix} I \downarrow D \uparrow \\ I \uparrow D \downarrow \end{matrix} \right)$

FURUKA is a normal product

d) Suppose, the price of CHOTPOTI,  $P_2$ , reduces to 30, while  $I$  &  $P_F$  remain at - 3 marks  
their initial values. Calculate XED.

$$Q_D = 200 - 3(50) + 0.5(5000) + 2(30)$$

$$\Rightarrow Q_D = 2610$$

$$XED = \frac{\frac{2610 - 2670}{\left[\frac{2610 + 2670}{2}\right]} \times 100\%}{\frac{30 - 60}{\left[\frac{30 + 60}{2}\right]} \times 100\%}$$

$$XED = 0.03$$

$$XED = \frac{\frac{\text{new } Q_{D1} - \text{old } Q_{D1}}{\left[\frac{\text{new } Q_{D1} + \text{old } Q_{D1}}{2}\right]} \times 100\%}{\frac{\text{new } P_2 - \text{old } P_2}{\left[\frac{\text{new } P_2 + \text{old } P_2}{2}\right]} \times 100\%}$$

$XED > 0$  SUBSTITUTES  
 $XED < 0$  COMPLEMENTARY

FURUKA & CHOTPOTI are substitutes of each other.

### Interpretations

1)  $|PED| > 1$  ELASTIC  
 $|PED| < 1$  INELASTIC

2)  $PES > 1$  ELASTIC  
 $PES < 1$  INELASTIC

3)  $YED > 0$  NORMAL  
 $YED < 0$  INFERIOR

4)  $XED > 0$  SUBSTITUTES  
 $XED < 0$  COMPLEMENTARY

$$PED - \text{price elasticity of demand} = \frac{\Delta Q_D}{\Delta P}$$

$$PES - \text{price elasticity of supply} = \frac{\Delta Q_S}{\Delta P}$$

$$YED - \text{income elasticity of demand} = \frac{\Delta Q_D}{\Delta I}$$

$$XED - \text{cross price elasticity of demand} = \frac{\Delta Q_{D1}}{\Delta P_2}$$